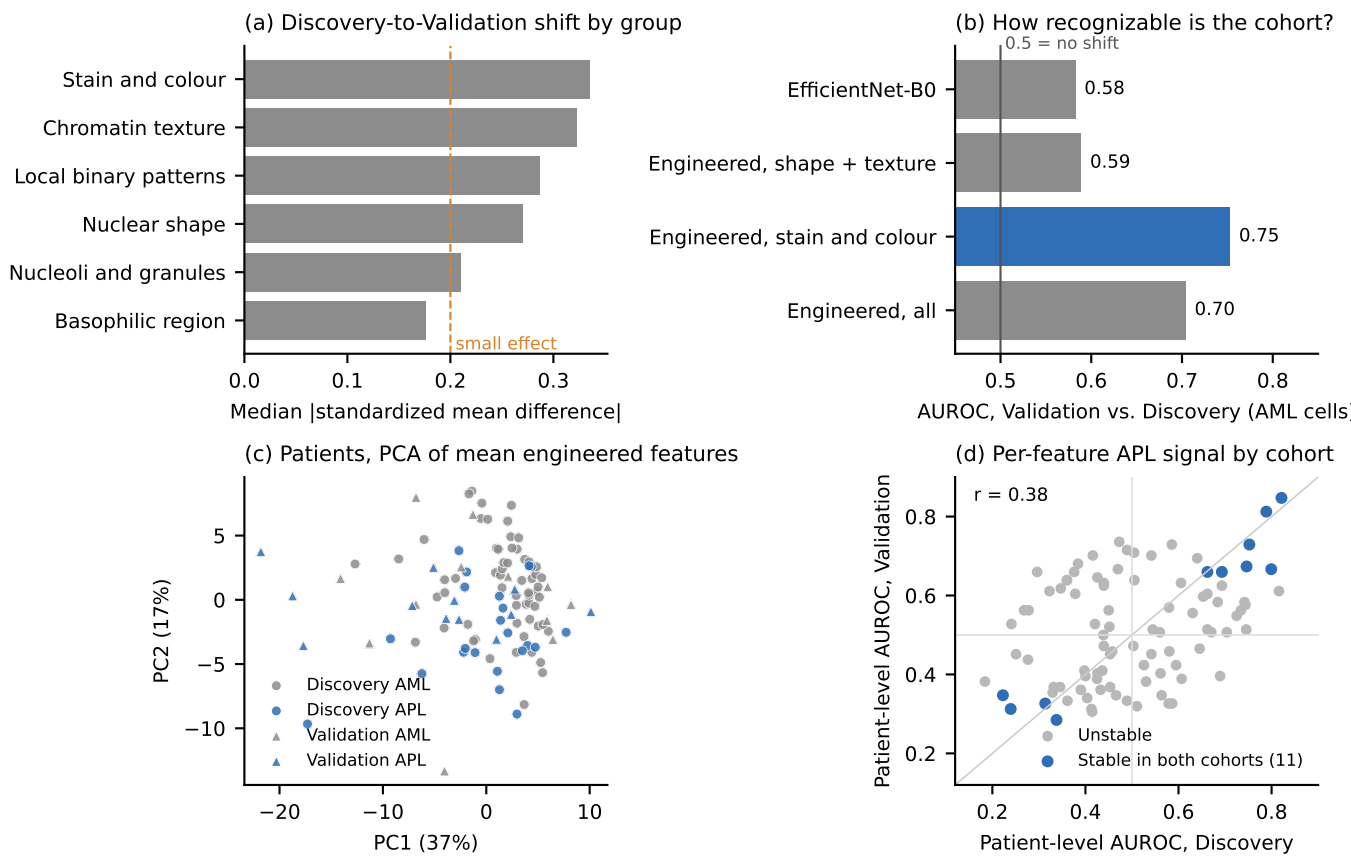
(a) Discovery-to-Validation shift by group
Stain and colour
Chromatin texture
Local binary patterns
Nuclear shape
Nucleoli and granules
Basophilic region
small effect
0.0
0.1
0.2
0.3
Median |standardized mean difference|
(b) How recognizable is the cohort?
0.5 = no shift
EfficientNet-B0
0.58
Engineered, shape + texture
0.59
Engineered, stain and colour
0.75
Engineered, all
0.70
0.5
0.6
0.7
0.8
AUROC, Validation vs. Discovery (AML cells)
(c) Patients, PCA of mean engineered features
PC2 (17%)
5
0
−5
−10
Discovery AML
Discovery APL
Validation AML
Validation APL
−20
−10
0
10
PC1 (37%)
(d) Per-feature APL signal by cohort
r = 0.38
Patient-level AUROC, Validation
0.8
0.6
0.4
0.2
Unstable
Stable in both cohorts (11)
0.2
0.4
0.6
0.8
Patient-level AUROC, Discovery